```
 874  0  INSERT FILE='E:/Documents/GitHub/biostatistics-portfolio/_spss_tmp/pbirths-regre
ssion.sps' ERROR=CONTINUE.
 874  0  * Set once: the folder that contains this repository, then run.
 875  0  * (CD set by runner)
 876  0  GET DATA
 877  0    /TYPE=XLS
 878  0    /FILE='data/raw/pbirths.xls'
 879  0    /SHEET=name 'Sheet1'
 880  0    /READNAMES=on.
 881  0  DATASET NAME DataSet1 WINDOW=FRONT.
 882  0  * Encoding: UTF-8.
 883  0
 884  0  * STEP 1: FITTING THE MODEL.
 885  0  * Go to: Analyze > Regression > Linear.
 886  0  * Move outcome variable to Dependent box.
 887  0  * Move all predictors to Independent(s) box using Shift-click.
 888  0  * Leave Method as "Enter" (all variables entered simultaneously).
 889  0  * ------------------------------------------------------------.
 890  0
 891  0  * ------------------------------------------------------------.
 892  0  * STEP 2: STATISTICS OPTIONS (click "Statistics" button).
 893  0  * Tick: Durbin-Watson       [tests independence of residuals].
 894  0  * Tick: Casewise diagnostics [identifies outliers >3 SD].
 895  0  * Tick: Collinearity diagnostics [gives VIF and Tolerance].
 896  0  * Tick: Part and partial correlations [zero-order = unadjusted correlations for
existence].
 897  0  * Tick: Confidence intervals if needed for final model reporting.
 898  0  * ------------------------------------------------------------.
 899  0
 900  0  * ------------------------------------------------------------.
 901  0  * STEP 3: PLOTS OPTIONS (click "Plots" button).
 902  0  * For LINEARITY and HOMOSCEDASTICITY (same plot tests both).
 903  0  * Y-axis: ZRESID (standardized residuals).
 904  0  * X-axis: ZPRED (standardized predicted values).
 905  0  * This is equivalent to a residuals-vs-fitted plot.
 906  0  * Tick: Produce all partial regression plots.
 907  0  * Under "Standardized Residual Plots" tick: Histogram and Normal probability plo
t.
 908  0  * These test NORMALITY of residuals.
 909  0  * ------------------------------------------------------------.
 910  0  DATASET ACTIVATE DataSet1.
 911  0  REGRESSION
 912  0    /MISSING LISTWISE
 913  0    /STATISTICS COEFF OUTS CI(95) R ANOVA COLLIN TOL ZPP
 914  0    /CRITERIA=PIN(.05) POUT(.10)
 915  0    /NOORIGIN
 916  0    /DEPENDENT grams
 917  0    /METHOD=ENTER black educ gestate smoke
 918  0    /SCATTERPLOT=(*ZRESID ,*ZPRED)
```

```
919  0    /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID)
920  0    /CASEWISE PLOT(ZRESID) OUTLIERS(3).
```

Regression

Notes

Output Created		03-NOV-2026 15:35:38
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS CI(95) R ANOVA COLLIN TOL ZPP /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT grams /METHOD=ENTER black educ gestate smoke /SCATTERPLOT=(*ZRESID ,*ZPRED) /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID) /CASEWISE PLOT(ZRESID) OUTLIERS(3).
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Warnings

Text: grams Command: REGRESSION An undefined variable name, or a scratch or system variable was specified in a variable list which accepts only standard variables. Check spelling and verify the existence of this variable. Execution of this command stops.
Text: black Command: REGRESSION An undefined variable name, or a scratch or system variable was specified in a variable list which accepts only standard variables. Check spelling and verify the existence of this variable.
Unrecognized text appears on the REGRESSION command. The only recognized subcommands are: Global options: DESCRIPTIVES MATRIX MISSING WIDTH; Case selection/weight: REGWGT SELECT; Variable list: VARIABLES; Equation options: CRITERIA NOORIGIN ORIGIN STATISTICS; Dependent variable(s): DEPENDENT; Equ. methods: METHOD BACKWARD ENTER FORWARD REMOVE STEPWISE TEST; Residuals: RESIDUAL CASEWISE PARTIALPLOT SAVE SCATTERPLOT OUTFILE TEMPLATE. Text found: gestate.
WARNING REGRESSION syntax scan continues. Further diagnostics from this command may be misleading - interpret with care.

```
 921  0
 922  0  * ------------------------------------------------------------.
 923  0  * STEP 4: INTERPRETING OUTPUTS.
 924  0
 925  0   * INDEPENDENCE.
 926  0  * Durbin-Watson statistic(1.915): acceptable range is 1 to 3.
 927  0  * A value within this range confirms independence of residuals.
 928  0
 929  0  * COLLINEARITY.(look at the Coefficients a table; last 2 columns for collinearit
y statitiscs)
 930  0  * VIF: no variable should exceed 10.
 931  0  * Tolerance: no variable should be below 0.1 (Tolerance = 1/VIF in SPSS).
 932  0  * Condition index and eigenvalues are provided but not routinely interpreted.
 933  0  * Condition   index >30 is flagged but commonly exceeded, especially with dummy
variables.
 934  0
 935  0  * EXISTENCE OF RELATIONSHIP.(look at the Coefficients a table; under correlation
s; specifically zero-order; equivalent of r
            unning a correlation matrix)
 936  0  * Zero-order correlations (unadjusted) should be at least 0.3 for key predictors
.
 937  0  * Can also judge from F-statistic significance in ANOVA table.
 938  0
```

```
939  0  * STRENGTH.
940  0  * SPSS reports standardized beta coefficients by default.
941  0  * The variable with the largest standardized beta has the strongest relationship
with the outcome.
942  0
943  0  * NORMALITY.
944  0  * Check the normal PP plot of residuals: points should lie on the diagonal.
945  0  * A histogram of standardized residuals should be approximately bell-shaped.
946  0
947  0
948  0  * ------------------------------------------------------------.
949  0
950  0  * ------------------------------------------------------------.
951  0
952  0  * STEP 5: SAVING RESIDUALS AND PREDICTED VALUES (click "Save" button).
953  0  * Save: Unstandardized predicted values and unstandardized residuals.
954  0  * Save: Standardized predicted values and standardized residuals.
955  0  * Save: Cook's Distance and Leverage (centered).
956  0  * Save: Standardized DFBeta(s).
957  0  * Saved variables appear automatically in the data editor.
958  0  REGRESSION
959  0    /MISSING LISTWISE
960  0    /CRITERIA=PIN(.05) POUT(.10)
961  0    /NOORIGIN
962  0    /DEPENDENT grams
963  0    /METHOD=ENTER black educ gestate smoke
964  0    /SCATTERPLOT=(*ZRESID ,*ZPRED)
965  0    /SAVE PRED ZPRED COOK ZRESID DFBETA SDBETA.
```

Regression

Notes

Output Created		03-NOV-2026 15:35:38
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT grams /METHOD=ENTER black educ gestate smoke /SCATTERPLOT=(*ZRESID ,*ZPRED) /SAVE PRED ZPRED COOK ZRESID DFBETA SDBETA.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.00

Warnings

Text: grams Command: REGRESSION An undefined variable name, or a scratch or system variable was specified in a variable list which accepts only standard variables. Check spelling and verify the existence of this variable. Execution of this command stops.
Text: black Command: REGRESSION An undefined variable name, or a scratch or system variable was specified in a variable list which accepts only standard variables. Check spelling and verify the existence of this variable.
Unrecognized text appears on the REGRESSION command. The only recognized subcommands are: Global options: DESCRIPTIVES MATRIX MISSING WIDTH; Case selection/weight: REGWGT SELECT; Variable list: VARIABLES; Equation options: CRITERIA NOORIGIN ORIGIN STATISTICS; Dependent variable(s): DEPENDENT; Equ. methods: METHOD BACKWARD ENTER FORWARD REMOVE STEPWISE TEST; Residuals: RESIDUAL CASEWISE PARTIALPLOT SAVE SCATTERPLOT OUTFILE TEMPLATE. Text found: gestate.
WARNING REGRESSION syntax scan continues. Further diagnostics from this command may be misleading - interpret with care.

```
 966  0
 967  0
 968  0  * ------------------------------------------------------------.
 969  0
 970  0  * ------------------------------------------------------------.
 971  0
 972  0  * OUTLIERS.
 973  0  * Casewise diagnostics lists cases with standardized residuals beyond ±3 SD(it l
ists the actaul observa tions as case numb
           ers).
 974  0  * Cook's Distance and Leverage are available in the residual statistics table.(t
o view these meaningfully we shall get the
           freq and the min and max checked in the stats)
 975  0  * Acceptable cutoffs: Cook's Distance <1; Leverage and standardized DFBetas <0.5
.
 976  0
 977  0
 978  0  FREQUENCIES VARIABLES=COO_1 SDB0_2 SDB1_2 SDB2_2 SDB3_2 SDB4_2
 979  0    /FORMAT=NOTABLE
 980  0    /STATISTICS=MINIMUM MAXIMUM
 981  0    /ORDER=ANALYSIS.
```

Frequencies

Notes

Output Created		03-NOV-2026 15:35:38
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=COO_1 SDB0_2 SDB1_2 SDB2_2 SDB3_2 SDB4_2 /FORMAT=NOTABLE /STATISTICS=MINIMUM MAXIMUM /ORDER=ANALYSIS.
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.03

Statistics

		Cook's Distance	Standardized DFBETA Intercept	Standardized DFBETA age	Standardized DFBETA weight	Standardized DFBETA height
N	Valid	250	250	250	250	250
	Missing	0	0	0	0	0
Minimum		.00000	-1.03138	-.30692	-1.26576	-.42852
Maximum		.45840	.59638	.32367	.62318	1.16900

Statistics

		Standardized DFBETA neck
N	Valid	250
	Missing	0
Minimum		-.33690
Maximum		.29136

```
 982  0
 983  0
 984  0  * STEP 6: MANUAL TEST FOR HOMOSCEDASTICITY (SPSS workaround).
 985  0  * SPSS has no built-in statistical test for equal variance (unlike Stata) .
 986  0  * Workaround: run a new regression with saved residuals as outcome.
 987  0  * and saved predicted values (or original predictors) as the predictor.
 988  0  * If equal variance holds, there should be NO significant relationship.
 989  0  * A p-value of ~1.0 (non-significant) confirms homoscedasticity is fulfilled.
 990  0  * This replicates what Stata does automatically with its residual-vs-fitted test
.
 991  0  * ------------------------------------------------------------.
 992  0  * ============================================================.
 993  0  * SPSS LINEAR REGRESSION: MODEL BUILDING (CONTINUED FROM PREVIOUS SESSION).
 994  0  * ============================================================.
 995  0
 996  0  * RECAP: Yesterday we confirmed all assumptions were fulfilled.
 997  0   * The workflow from here: assumptions  interaction  confounding  final mode
l.
 998  0
 999  0  * ------------------------------------------------------------.
1000  0  * STEP 7: GENERATING INTERACTION TERMS.
1001  0  * SPSS linear regression CANNOT generate interaction terms automatically.
1002  0  * (Unlike logistic regression, which can generate them within the dialog box).
1003  0  * You must create interaction terms manually before running the model.
1004  0  * Go to: Transform > Compute Variable.
1005  0  * Multiply the relevant variables together (e.g., black*smoke = blacksmoke).
1006  0  * Generate all candidate interaction terms this way, one at a time.
1007  0  * Run all compute statements from the syntax editor at once for efficiency.
1008  0  * ------------------------------------------------------------.
1009  0
1010  0  * ------------------------------------------------------------.
1011  0  * STEP 8: TESTING FOR INTERACTION — NO MAIN EXPOSURE SCENARIO.
1012  0  * (Research question: "What factors are associated with birthweight?").
1013  0
1014  0  * When there is no designated main exposure:.
1015  0  * - Form all possible interaction terms between variables if sample size allows.
1016  0  * - Rule of thumb: need at least 10 observations per variable (including each du
mmy).
```

```
1017  0  * - Count dummy variables separately; a 3-category variable = 2 variables.
1018  0  * - If sample size is insufficient, first identify significant factors via stepw
ise,.
1019  0  *   then form interaction terms only between those significant factors.
1020  0
1021  0  * To run the model: Analyze > Regression > Linear.
1022  0  * Enter the dependent variable, then all basic (main) variables and interaction
terms.
1023  0  * Set Method to BACKWARD.
1024  0  * (Backward is preferred because it handles negative confounding better than for
ward:.
1025  0  *  starting with all variables together allows masking effects to be detected).
1026  0  * Default significance level of 5% is acceptable; leave Options as they are.
1027  0  * Click OK or Paste to Syntax.
1028  0  * ------------------------------------------------------------.
1029  0
1030  0  * ------------------------------------------------------------.
1031  0  * STEP 9: READING THE BACKWARD ELIMINATION OUTPUT.
1032  0  * The model summary table shows all steps (models) fitted.
1033  0  * Each step shows the R-squared, adjusted R-squared, and predictors remaining.
1034  0  * Focus on the LAST STEP — this is the final interaction model.
1035  0  * Check which interaction terms have survived.
1036  0
1037  0  * CRITICAL RULE: Before accepting a surviving interaction term as valid,.
1038  0  *   confirm that ALL its component (basic) variables are present in the model.
1039  0  * If a basic variable was dropped (e.g., education was dropped but black*educati
on survived):.
1040  0  *   - Force that basic variable back in.
1041  0  *   - Change Method to ENTER and re-run.
1042  0  *   - Check whether the interaction term remains significant.
1043  0  *   - If it loses significance: it was a proxy, not a true interaction — drop i
t.
1044  0  *   - If it stays significant: retain both the basic variable and the interactio
n term.
1045  0  * ------------------------------------------------------------.
1046  0
1047  0  * ------------------------------------------------------------.
1048  0  * STEP 10: TESTING FOR INTERACTION — WITH A MAIN EXPOSURE.
1049  0  * (Research question: "Does smoking affect birthweight?").
1050  0
1051  0  * When there IS a designated main exposure (e.g., smoking):.
1052  0  * - Form interaction terms ONLY between the main exposure and all other variable
s.
1053  0  * - Basic variables must all remain in the model (you cannot exclude them).
1054  0  * - This is because higher-order terms (interaction terms) cannot be in a model.
1055  0  *   without their component basic variables — this is a fundamental modelling r
ule.
1056  0  * - Set Method to BACKWARD and run.
1057  0  * - Focus on the last step to identify which smoke* interaction terms survive.
```

```
1058  0  * - Verify that all basic variables of surviving interaction terms are present.
1059  0  * ------------------------------------------------------------.
1060  0
1061  0  * ------------------------------------------------------------.
1062  0  * STEP 11: ASSESSING FOR CONFOUNDING.
1063  0  * Confounding is assessed for the MAIN EXPOSURE only (e.g., smoking coefficient)
.
1064  0  * The starting model for confounding assessment includes:.
1065  0  *   - All basic variables.
1066  0  *   - Only the significant interaction terms from Step 10.
1067  0  *   - Variables that held significant interaction terms must remain, even if not
 significant.
1068  0
1069  0  * Procedure (using BACKWARD method):.
1070  0  * Note which basic variables are dropped by the backward procedure.
1071  0  * Bring them back one at a time, starting with the LAST variable to be dropped.
1072  0  * (The last dropped is assessed first; work backwards through the order of dropp
ing).
1073  0
1074  0  * For each variable assessed for confounding:.
1075  0  * Run ENTER method with that variable included.
1076  0  * Compare the main exposure coefficient (e.g., smoke) to the model without it.
1077  0  * If the coefficient changes by 10% or more: it IS a confounder — keep it.
1078  0  *   That model then becomes the reference (crude) model for the next variable.
1079  0  * If the coefficient changes by less than 10%: it is NOT a confounder — drop it
.
1080  0  *   The previous reference model remains the crude model for the next variable.
1081  0  * ------------------------------------------------------------.
1082  0
1083  0  * ------------------------------------------------------------.
1084  0  * STEP 12: STRATIFIED ANALYSIS (OPTIONAL BUT RECOMMENDED WITH INTERACTION).
1085  0  * When a significant interaction term is present, stratified analysis helps.
1086  0  *   interpret what is actually happening in each group.
1087  0  * In SPSS, use Data > Select Cases to filter by the stratifying variable.
1088  0  * Run the regression model separately for each stratum.
1089  0  * Example: to explore smoking*black interaction,.
1090  0  *   run one model for non-blacks (black=0) and one for blacks (black=1).
1091  0  *   Or stratify by smoking status to see the effect of race in each group.
1092  0  * Stratification is useful for interpretation and reporting of interaction effec
ts.
1093  0  * It is not required for the Stata analysis but applies equally to both software
.
1094  0  * ------------------------------------------------------------.
1095  0
1096  0  * ------------------------------------------------------------.
1097  0  * STEP 13: FINAL MODEL AND REPORTING.
1098  0  * The final model contains: significant basic variables + significant interactio
n terms + confirmed confounders.
1099  0  * Tick Confidence Intervals under Statistics before the final run.
```

```
1100  0  * Report: regression coefficients, 95% CIs, p-values, R-squared, adjusted R-squa
red.
1101  0  * Interpret coefficients as done in the Stata session (previous notes).
1102  0  * When a basic variable is non-significant but its interaction term is significa
nt:.
1103  0  *   it means that variable has no effect in the reference group,.
1104  0  *   but its effect differs significantly in the interacting group — keep it.
1105  0
1106  0  * IF AN ASSUMPTION IS VIOLATED:.
1107  0  * - Normality violation: attempt transformation (e.g., log), then re-check all a
ssumptions.
1108  0  *   (Transforming for normality can affect homoscedasticity — verify both).
1109  0  * - If transformation fails: consider a different regression model altogether.
1110  0  * ------------------------------------------------------------.
1111  0
1112  0  * End of INSERT and INCLUDE nesting level 02.
1112  0
1113  0  OUTPUT EXPORT /CONTENTS EXPORT=ALL /PDF DOCUMENTFILE='E:/Documents/GitHub/biosta
tistics-portfolio/linear-regression/spss/sp
            ss-files/pbirths-regression.pdf'.
```